

9. This problem has to do with *odds*.

- (a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

$$\text{odds} = \frac{p}{1-p}$$

(a)

$$0.37 = \frac{p}{1-p}$$

$$0.37(1-p) = p$$

$$p = \frac{0.37}{1.37} \quad p \approx 0.27 = 27\% \#$$

- (b) Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?

$$p = 0.16$$

$$\text{odds} = \frac{0.16}{1-0.16} = \frac{0.16}{0.84} \approx 0.19 \#$$